

# Mark Christian Messner

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## Argonne National Laboratory

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## Positions

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2023-

### **Group Leader**, *Argonne National Laboratory*

Established the Thermal and Structural Materials Modeling and Simulations Group.

Led a team of 3 staff and 4-6 postdocs on projects funded by the U.S. Department of Energy, Office of Nuclear Energy, the Office of Energy Efficiency and Renewable Energy, the U.S. Nuclear Regulatory Agency, and other sponsors.

Managed a research portfolio with greater than \$3 million per year in total funding. Established new research portfolios in modeling and simulation for advanced manufacturing processes and the reliability of monolithic ceramics and ceramic matrix composites.

Recognized as a national expert in high temperature design.

2016-

### **Principal Mechanical Engineer**, *Argonne National Laboratory*

Research topics: High temperature structural materials, design of high temperature nuclear reactors and concentrating solar power systems, crystal plasticity, machine learning methods for materials and material constitutive modeling, qualification of AM nuclear components

Led work on the revision and improvement of several parts of the American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section III, Division 5 covering the design and construction of high temperature nuclear reactors

2014-2016

### **Postdoctoral Researcher**, *Lawrence Livermore National Laboratory*

Supervisor: Nathan Barton

Research topics: Multiscale material modeling of additively manufactured structured materials, modeling and optimization of lattice-structured meta-materials, multiscale modeling of HCP metals

2010-2014

### **Research Assistant**, *University of Illinois at Urbana-Champaign*

Supervisor: Robert Dodds, Jr.

Research topics: Parallel performance of WARP3D, crystal plasticity, mesoscale modeling of fatigue/fracture processes, homogenization and multiscale damage calculations

## Education

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2011-2014

### **Doctor of Philosophy**, *University of Illinois at Urbana-Champaign*

GPA 3.96/4.00

Major: Civil and Environmental Engineering

Advisor: Robert Dodds, Jr.

Dissertation: *Micromechanical models of delamination in Al-Li alloys*

Computational Science and Engineering Certificate

2010-2011

### **Master of Science**, *University of Illinois at Urbana-Champaign*

GPA 3.96/4.00

Major: Civil and Environmental Engineering  
 Advisor: Robert Dodds, Jr.  
 Computational Science and Engineering Certificate  
 2006-2010 **Bachelor of Science**, *University of Illinois at Urbana-Champaign*  
 GPA 3.97/4.00  
 Major: Civil and Environmental Engineering, Minor: German  
 Degree awarded with Highest Honors and University Honors

## Honors/Awards

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| 2025             | ASME G.E.O. Widera Literature Award for Authoring the Outstanding Technical Paper in the Journal of Pressure Vessel Technology                |
| 2023             | Doug Scarth Early Career Leadership Award for Outstanding Service to the Pressure Vessels & Piping (PVP) Division as an Early Career Engineer |
| 2022             | Secretary of Energy Achievement Award                                                                                                         |
| 2022, 2023, 2025 | ASME Boiler & Pressure Vessel Code Certificate of Acclamation                                                                                 |
| 2020             | Impact Argonne Award for Innovation                                                                                                           |
| 2012-14          | National Defense Science & Engineering Graduate Fellowship                                                                                    |
| 2010-11          | University Fellowship                                                                                                                         |
| 2008-            | Tau Beta Pi                                                                                                                                   |

## Professional Affiliations

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| 2016-     | ASME      |
| 2016-     | ANS       |
| 2014-2016 | APS       |
| 2006-2014 | ASCE, EMI |

## Professional Service

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| 2018-     | Generation IV Forum: Task Group/Working Group on Advanced Manufacturing and Materials Engineering <i>co-chair</i>                                                                                                                                                                                                                                                                  |
| 2020-     | ASME Boiler & Pressure Vessel Code <i>committee chair</i> : BPV III WG on Analysis Methods                                                                                                                                                                                                                                                                                         |
| 2018-2020 | ASME Boiler & Pressure Vessel Code <i>committee chair</i> : BPV III SWG on Inelastic Analysis Methods                                                                                                                                                                                                                                                                              |
| 2017-     | ASME Boiler & Pressure Vessel Code <i>committee member</i> : BPV III SG High Temperature Reactors, WG on Analysis Methods, WG High Temperature Flaw Evaluation, WG Creep-Fatigue and Negligible Creep; BPTCS/BNCS Special Committee on Use of Additive Manufacturing for Pressure Retaining Equipment, BVP I/VIII WG on Elevated Temperature Design, and several other committees. |
| 2017-     | PVP Conference: Technical Program Representative, co-Technical Program Representative, Honors and Awards Chair, Materials and Fabrication Secretary and Vice Chair, track co-chair                                                                                                                                                                                                 |
| 2018-2019 | WCCM/USNCCM: <i>track organizer</i>                                                                                                                                                                                                                                                                                                                                                |

## Other Skills and Qualifications

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| Languages: | German (Proficient) |
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